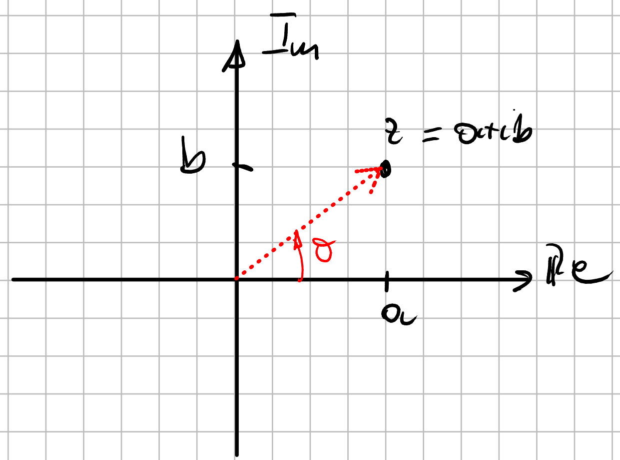
Sea $z \in \mathbb{C}$ → $z = a + ib$ con a y $b \in \mathbb{R}$ → Forma rectangular de un numero complejo

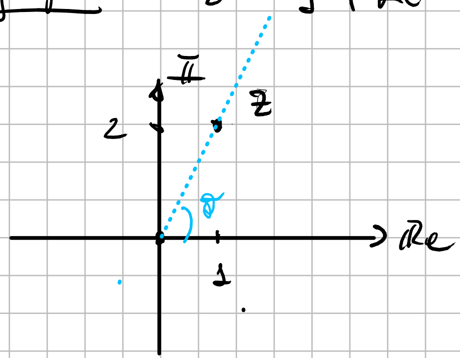
$Re(z) = a \quad Im(z) = b$



en forma polar $z = |z| \, e^{i\theta}$ donde $|z| = \sqrt{a^2 + b^2}$ (modulo)

$\theta = Arg(z)$ (función argumento)

<u>Ejemplo</u> $z = 1 + 2i$



$|z| = \sqrt{1^2 + 2^2} = \sqrt{5}$

$tg(\theta) = \frac{2}{1} \Rightarrow \theta = Arctg(2)$

$z = \sqrt{5} \, e^{i \, Arctg(2)}$

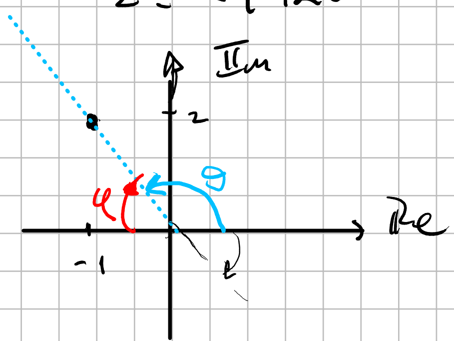
Recordar SOHCAHTOA

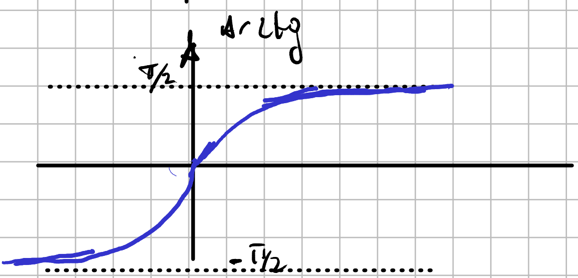
$z = -1 + 2i$

$|z| = \sqrt{(-1)^2 + (2)^2} = \sqrt{5}$

$\theta = \pi - \varphi = \pi - Arctg\left(2/-1\right) =$

$\varphi = Arctg\left(2/-1\right) = -Arctg(2)$

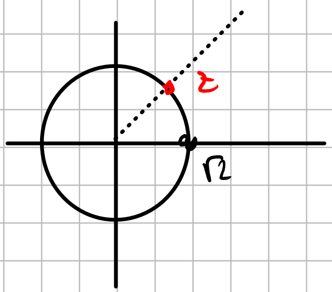




formula de euler

$$e^{i\theta} = \cos(\theta) + i\,\mathrm{sen}(\theta)$$

$$z = \sqrt{2}\, e^{i\pi/4} = \underbrace{\sqrt{2}\left[\cos(\pi/4) + i\,\mathrm{sen}(\pi/4)\right]}_{\text{Formula trigonometrica.}} = \sqrt{2}\left[\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right] = \underbrace{1+i}_{\text{Forma rectangular}}$$



Ejemplo

1) b) ii) $\quad p = z_1^3 z_2^3 \qquad z_1 = i \qquad z_2 = 1+i$

$$z_1^3 = i\,i\,i = -i$$

$$z_2 = (1+i)^3 = (1+i)(1+i)(1+i) = \left(1 + 2\cdot 1\cdot i + (i)^2\right)(1+i)$$
$$= (1+2i-1)(1+i) = 2i-2$$

suponer

$$z = a+ib = |z|\, e^{i\varphi}$$

$$z^n = (\underbrace{|z|}_{\in \mathbb{R}}\, e^{i\varphi})^n = |z|^n (e^{i\varphi})^n = |z|^n e^{in\varphi}$$

$$z = 1+i = \sqrt{2}\, e^{i\pi/4}$$

$$z^3 = \left((\sqrt{2})^3\, e^{i\frac{3\pi}{4}}\right) = 2\sqrt{2}\left(\cos\left(\tfrac{3\pi}{4}\right) + i\,\mathrm{sen}\left(\tfrac{3\pi}{4}\right)\right)$$

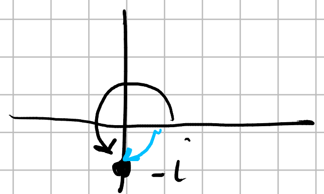
$$= \cancel{2}\sqrt{2}\left(-\frac{\sqrt{2}}{\cancel{2}} + i\frac{\sqrt{2}}{\cancel{2}}\right) = -2 + i2.$$

$$z_1^3 \cdot z_2^3 = (-i)(-2+2i) = +2i+2$$

Nota

$$z_1 = a+ib = |z_1|\, e^{i\theta_1}$$
$$z_2 = \alpha + i\beta = |z_2|\, e^{i\theta_2} \Rightarrow z_1 z_2 = \left(|z_1|\, e^{i\theta_1} \cdot |z_2|\, e^{i\theta_2}\right)$$
$$= |z_1|\,|z_2|\, e^{i(\theta_1+\theta_2)}$$

$$z_1^3 \cdot z_2^3 = (-i)(-2+2i) = \underbrace{1\, e^{i\frac{3\pi}{2}}}_{=1\cdot e^{-i\frac{\pi}{2}}} \cdot 2\sqrt{2}\, e^{i\frac{3\pi}{4}}$$



$$= 2\sqrt{2}\, e^{i\frac{9\pi}{4}} = 2\sqrt{2}\left[\cos\left(\frac{9\pi}{4}\right) + i \operatorname{sen}\left(\frac{9\pi}{4}\right)\right]$$

$$= 2\sqrt{2} \cdot \left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = 2+2i$$

Ejemplo: resolver la siguiente ec con $w \in \mathbb{C}$

$$w^4 = i \quad (\text{similar a buscar } \sqrt[4]{i})$$

$$w = a+ib \Rightarrow (a+ib)^4 = i \quad \times$$

$$w = \rho e^{i\theta} \Rightarrow w^4 = (\rho e^{i\theta})^4 = \rho^4 e^{i4\theta}$$

$$\Rightarrow \rho^4 e^{i4\theta} = i = 1\, e^{i\pi/2}$$

$$\begin{cases} \rho^4 = 1 \\ 4\theta = \pi/2 + 2k\pi \end{cases} \Rightarrow \quad \rho = \sqrt[4]{1} = 1, \quad \theta = \frac{\pi/2 + 2k\pi}{4} = \frac{\pi}{8} + \frac{k\pi}{2}$$

con $k \in \mathbb{Z}$

$$w_k = 1\, e^{i\left(\frac{\pi}{8}+\frac{k 5\pi}{2}\right)} = 1\, e^{i\left(\frac{\pi + 4k\pi}{8}\right)}$$

$$\begin{cases} w_0 = 1\, e^{i\,\pi/8} \\ w_1 = 1\, e^{i\,5\pi/8} \\ w_2 = 1\, e^{i\,9\pi/8} \\ w_3 = 1\, e^{i\,13\pi/8} \end{cases}$$

Sol: $\{ w_0 \cdots w_3 \}$

$$w_4 = 1\, e^{i\,\frac{\pi+16\pi}{8}} = e^{i\left(\frac{\pi}{8}+\frac{16\pi}{8}\right)} = e^{i\left(\frac{\pi}{8}+2\pi\right)} = e^{i\frac{\pi}{8}}\, \underbrace{e^{i2\pi}}_{=1}$$

<u>AUX</u>

$$e^{i2\pi} = \cos(2\pi) + i\,\text{sen}(2\pi) = 1$$

$\therefore\ w_4 = w_0$

$w_5 = w_1$

$w_6 = w_2$

$$w^4 = i \Rightarrow w^4 - i = 0$$

—o—o—

raíz n-esimo de un $z \in \mathbb{C}$

Sea $z \in \mathbb{C}$ de la forma $z = \rho\, e^{i\theta}$

$$\boxed{\sqrt[n]{z} = \sqrt[n]{\rho}\ e^{i\left(\frac{\theta}{n}+\frac{2k\pi}{n}\right)} \quad \text{con } k \in \{0, 1, \cdots, n-1\}}$$

2. d) $\underbrace{x + 2y + (4x-3y)\,i}$ $= \underbrace{2x - 1 + (y-6)\,i}$

$$\begin{cases} x + 2y = 2x - 1 \\ 4x - 3y = y - 6 \end{cases}$$

$$\begin{cases} -x + 2y = -1 \\ 4x - 4y = -6 \end{cases} \Rightarrow \begin{cases} x = 1 + 2y \\ 4 + 8y - 4y = -6 \end{cases} \Rightarrow$$

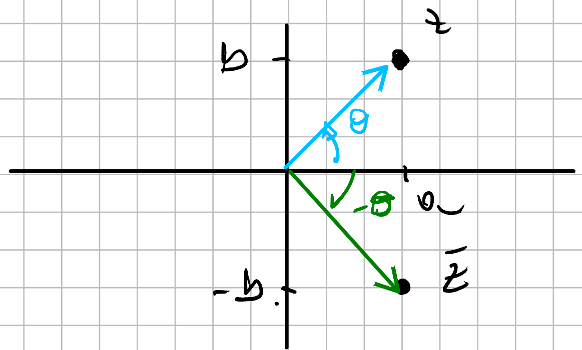
$$4y = -10 \Rightarrow y = -\frac{5}{2}$$

$$\begin{bmatrix} -1 & 2 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ -6 \end{bmatrix}$$

4) $z^n = \bar{z}$ $\qquad$ $z = \rho e^{i\vartheta}$

$$z = a + ib \rightarrow \bar{z} = a - ib = |z| e^{i(-\vartheta)}$$

$$|z| = |\bar{z}|$$



$$z^n = \bar{z} \Rightarrow \rho^n e^{in\vartheta} = \rho\, e^{-i\vartheta}$$

$$\begin{cases} \rho^n = \rho \Rightarrow \rho^n - \rho = 0 \Rightarrow \rho(\rho^{n-1} - 1) = 0 \\ n\vartheta = -\vartheta + 2k\pi \end{cases}$$

$$\rho(\rho - 1)\underbrace{p(\rho)}_{\text{polinomio de grado 2}} = 0 \Rightarrow \boxed{\rho = 1}$$

$$n\vartheta + \vartheta = 2k\pi$$

$$\vartheta(n+1) = 2k\pi \Rightarrow \vartheta = \frac{2k\pi}{n+1}$$

$$z = 1\, e^{i\frac{2k\pi}{n+1}} \quad \text{con} \quad k \in \{0 \dots n\}$$

5) si $z = a + ib$

$$z + \bar{z} = a + \cancel{ib} + a - \cancel{ib} = 2a = 2\,\mathrm{Re}\{z\}$$

$$\boxed{\mathrm{Re}\{z\} = \frac{z+\bar{z}}{2}}$$

$$z - \bar{z} = \cancel{a} + ib - \cancel{a} + ib = 2ib = 2i\,\mathrm{Im}(z)$$

$$\boxed{\mathrm{Im}(z) = \frac{z-\bar{z}}{2i}}$$

8k) $z^2 + |z^2| = i\bar{z}$ si $z = a + ib = \rho\, e^{i\theta}$

$$z^2 = (a+bi)^2 = a^2 - b^2 + 2iab = a^2 - b^2 + i\,2ab$$

$$\mathrm{Re}\{z^2\} = a^2 - b^2$$

$$\mathrm{Im}\{z^2\} = 2ab$$

$$|z^2| = \sqrt{\underbrace{(a^2-b^2)^2}_{a^4 - 2a^2b^2 + b^4} + (2ab)^2} = \sqrt{\underbrace{a^4 + 2a^2b^2 + b^4}_{(a^2+b^2)^2}}$$

$$= a^2 + b^2$$

$$z^2 = \rho^2 e^{i2\theta} \Rightarrow |z^2| = \rho^2 = \left(\sqrt{a^2+b^2}\right)^2$$

$$= a^2 + b^2$$

$$z^2 + |z^2| = i\bar{z}$$

$$a^2 \cancel{-b^2} + i2ab + a^2 \cancel{+b^2} = i(a-ib) = b+ia$$

$$\begin{cases} 2a^2 = b \\ 2ab = a \end{cases} \Rightarrow 2a(2a^2) = a$$

$$2a^3 - a = 0$$

10) $\bar{z} = z^3 (z-i)^4 \quad \wedge \quad |z| = 1$

$$z\bar{z} = z^4 (z-i)^4 \Rightarrow 1 = z^4 (z-i)^4$$

CAUX

$z = x+iy$

$\bar{z} = x-iy$

$$\Rightarrow z\bar{z} = (x+iy)(x-iy) = x^2 + \cancel{ixy} - \cancel{iyx} + y^2$$

$$= x^2+y^2 = |z|^2$$

$$\left.\begin{array}{l} 1 = z^4 (z-i)^4 = [\bar{z}(z-i)]^4 \\ w = z(z-i) \end{array}\right\} \Rightarrow 1 = w^4$$

$$w^4 - 1 = 0$$

$$(w^2-1)(w^2+1) = 0$$

$$(w+1)(w-1)(w^2+1) = 0$$

$w = 1$

$w = -1$

$w = i$

$w = -i$

$$z(z+i) = 1 \Rightarrow z^2 + iz - 1 = 0$$

11 a) $e^{w} = z \qquad z = \sqrt{3} - i\sqrt{3}$

$e^{w} = \sqrt{3} - i\sqrt{3}$ con $w \in \mathbb{C}$ $\therefore$ $w = \alpha + i\beta = \rho e^{i\theta}$

$\left(\cancel{e^{\rho e^{i\theta}}}\right)$

$$\underbrace{e^{\alpha + i\beta}} = \sqrt{3} - i\sqrt{3} \Rightarrow \underbrace{e^{\alpha} e^{i\beta}}_{\text{numero complejo}} = \overbrace{\sqrt{3} - i\sqrt{3}}^{\text{Forma rectangular}}$$

$$\underbrace{e^{\alpha}}_{\text{modulo}} e^{i\beta} \xrightarrow{} \text{Argumento}$$

numero complejo expresado en forma polar de modulo e^{α} y argumento β

$$|\sqrt{3}\ i\sqrt{3}| = \sqrt{(\sqrt{3})^2 + (\sqrt{3})^2} = \sqrt{6}$$

$$\operatorname{Arg}(\sqrt{3} + i\sqrt{3}) = \operatorname{Arctg}(\sqrt{3}/\sqrt{3}) = \operatorname{Arctg}(1) = \pi/4$$

$$\boxed{e^{\alpha} e^{i\beta} = \sqrt{6}\, e^{i\pi/4}} \Rightarrow \begin{cases} e^{\alpha} = \sqrt{6} \Rightarrow \alpha = \ln(\sqrt{6}) \\ \beta = \pi/4 + 2k\pi \quad \text{con } k \in \mathbb{Z} \end{cases}$$

$$\boxed{w_k = \ln(\sqrt{6}) + i\ \frac{\pi}{4} + 2k\pi \qquad \text{con } k \in \mathbb{Z}}$$